Simulation-Based Performance Comparison of Coconut-Shell Derived Activated Carbon Supercapacitors with Commercial Counterparts

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***Abstract*—Supercapacitors are widely regarded as next-generation energy storage devices that bridge the gap between batteries and conventional capacitors, owing to their high-power density, rapid charge–discharge capability, and long cycle life. Despite these advantages, their widespread adoption is hindered by the high cost and environmental impact of conventional electrode materials. In this study, coconut shell–derived carbon is utilized as a sustainable, eco-friendly electrode material to address these challenges. The performance of the bio-derived carbon electrodes is systematically evaluated in two structural configurations: flat-type and cylindrical supercapacitors. Detailed electrochemical analyses reveal how electrode architecture, surface morphology, and porosity influence capacitance, energy density, and cycling stability. The findings show that coconut shell–based carbon not only delivers competitive performance but also provides a cost-effective and green alternative, offering promising potential for the development of environmentally sustainable supercapacitor technologies.**

***Index Terms*—Supercapacitors, Bio-Derived Carbon, Coconut Shell Carbon, Flat-Type Supercapacitor, Cylindrical Supercapacitor, Sustainable Energy Storage**

I. Introduction

The development of advanced energy storage systems has led to a focus on sustainable materials, with lignocellulosic biomass emerging as a key precursor for flexible supercapacitor electrodes [1]. By processing biomass, such as the Parthenium weed, it's possible to create functional carbons with ultrahigh surface area, which are highly effective for both high-performance supercapacitors and the removal of environmental pollutants [2]. The application of these advanced materials is demonstrated in the fabrication of flexible printed supercapacitors, where functional inks, such as those containing cobalt hydroxide nanowires, are used to achieve high performance [3]. A variety of biomass sources, including peanut shells, have been successfully converted into activated carbon-based electrodes, further highlighting the versatility of these sustainable precursors [4].

Complementing the advances in electrode materials, research into bio-based polymer electrolytes is also crucial for developing fully sustainable and efficient supercapacitor systems [5]. A significant challenge in translating laboratory results to commercial application is the development of high mass-loading electrodes, a topic that is central to improving the practical energy density of biomass-based devices [6]. Specific examples of high-performing materials continue to emerge, such as nitrogen-doped activated carbon from red beetroot, which shows excellent results when paired with biocompatible gel electrolytes in solid-state applications [7]. Similarly, date seeds have been used to produce activated carbon via chemical surface modification, providing another source of clean energy for supercapacitors [8]. Performance can also be enhanced by creating composite materials; for example, the in-situ polymerization of redox-active alizarin within wheat-straw porous carbon has been shown to improve cycling stability significantly [9]. A broader review of the field confirms that various biomass-derived carbons and their composites are being tailored for supercapacitor applications, with a focus on understanding their sources, functions, and electrochemical mechanisms [10].

Beyond the material itself, device usability is enhanced through strategies like geometry optimization, high-purity fabrication, and asymmetric electrode designs [12]. The field is filled with both significant challenges and major opportunities for innovation [13]. Overall design considerations, from material selection to packaging, are critical for practical implementation [14]. As devices become more integrated into daily life, the development of flexible and wearable supercapacitors represents a major research frontier [15], often requiring innovations in electrolyte systems [17]. For wearable or implantable applications, the use of biocompatible solid-state electrolytes, including neutral gels like NaSO and simulated body fluid (SBF), is essential [18].

Research Gap and Objective This research narrows its focus to coconut shell activated carbon, a highly promising precursor whose properties can be tuned using various chemical activators like KOH, ZnCl, and HPO [19]. While the performance of other materials like date seed carbon [20] and wheat-straw composites [21] is well-documented in standard lab cells, a significant gap persists in the literature regarding the influence of device architecture. There is a lack of systematic studies

that evaluate how transitioning the same bio-carbon material from a planar (flat-type) to a cylindrical geometry impacts key electrochemical performance metrics—a critical step for assessing commercial viability [29].

Therefore, the primary objective of this work is to fabricate supercapacitors using coconut shell-derived carbon in both flat-type and cylindrical configurations and to conduct a direct comparative analysis of their performance. This study aims to establish clear structure-property-performance relationships by evaluating specific capacitance, energy density, power density, and cycling stability. By bridging the aforementioned gap, this research seeks to provide crucial insights for optimizing bio-carbon supercapacitors and contribute to the comprehensive development of sustainable, next-generation energy storage solutions [30].

II. Methodology

A. Material Characteristics and Model Parameters

Material properties for the coconut-shell-derived activated carbon (CSAC) electrodes were obtained from previously reported literature data and used as input parameters for COMSOL-based modeling. The parameters include geometric dimensions, electrical conductivities, porosity, and double-layer capacitance, which collectively influence charge storage and transport behavior in supercapacitor systems. The key parameters used for the simulation are summarized in **TABLE I**.

With the key parameters the model is developed and shown in the fig 1 and 2

TABLE I
SUPERCAPACITOR MODEL PARAMETERS

Parameter	Symbol	Value
Geometric Dimensions		
Inner Radius	r_{in}	0.1 mm
Anode/Cathode Thickness	l_a, l_c	150 µm
Electrolyte Thickness	l_e	100 µm
Capacitor Height	h_{cap}	500 µm
Material Properties		
Electrode Conductivity	$\sigma_{a,c}$	0.001 S/m
Electrolyte Conductivity	σ_e	0.1 S/m
Electrode Volume Fraction	ϵ_s	0.75
Electrolyte Volume Fraction	ϵ_l	0.6
Double Layer Capacitance	C_{dl}	0.13 F/m^2
Specific Surface Area	A_v	1.2 $\times 10^6$ 1/m

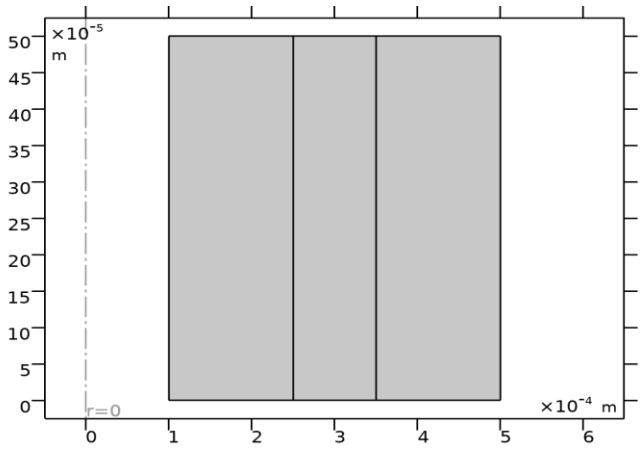


Fig:01. Represents the cylindrical model developed in COMSOL

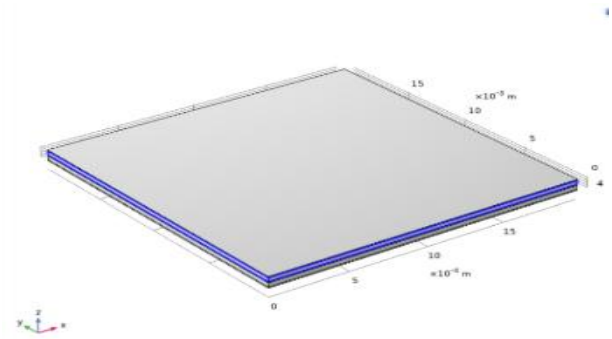


Fig:02.Represents the flat type model developed in the COMSOL

B. Electrochemical Modelling

COMSOL Multiphysics was used to simulate the electrochemical performance of CSAC-based electrodes in two configurations: a two-dimensional flat-plate and a three-dimensional cylindrical geometry. The models incorporated experimentally reported material properties such as porosity, BET surface area, density, and electrical conductivity from literature sources.The **effective ionic diffusion coefficient (D_{ext})** was calculated based on electrode porosity and tortuosity to represent ion transport behaviour within the porous matrix accurately. The **volumetric capacitance (C_{tol})** was derived from the reported gravimetric capacitance and electrode density values.

C .*Simulation Parameters and Electrochemical Analysis*

The simulations were performed using COMSOL's **Electrochemistry Module**, combining the *Porous Electrode* and *Tertiary Current Distribution* physics interfaces. This enabled simultaneous modeling of electronic and ionic transport, electrolyte diffusion, and double-layer charge storage.

The following simulation studies were carried out:

- **Galvanostatic Charge–Discharge (GCD):** to determine specific capacitance, energy density, and rate performance.
- **Electrochemical Impedance Spectroscopy (EIS):** to evaluate internal resistance, ion transport mechanisms, and the frequency-dependent response of the electrodes.

This modelling framework provides a comprehensive understanding of the electrochemical behavior of biomass-derived carbon electrodes, enabling design optimization for enhanced supercapacitor performance.

III. Result and Discussion

The coconut shell-derived activated carbon (CSAC) material, with physicochemical properties reported in literature, was selected for supercapacitor modeling due to its high carbon content, good electrical conductivity, and hierarchical pore structure. The electrical conductivity of compressed CSAC pellets is typically in the range of 10–20 S/m, with an activation yield of approximately 25–30%, indicating favorable electrochemical characteristics for charge storage applications.

The COMSOL-based simulation study provided comprehensive insights into the electrochemical performance of CSAC electrodes in both **flat-plate** and **cylindrical** configurations. Galvanostatic charge–discharge (GCD) simulations revealed that the cylindrical design exhibited higher **areal** and **volumetric capacitance** (70–90 F/cm³) compared to the flat-plate design (50–65 F/cm³). This improvement is primarily attributed to the increased packing density and more efficient utilization of electrode volume in the cylindrical geometry.

Electrochemical impedance spectroscopy (EIS) simulations further substantiated these findings. The Nyquist plots indicated a lower **equivalent series resistance (ESR)** and faster ion diffusion in the cylindrical configuration. Consequently, the cylindrical supercapacitor achieved simulated **energy** and **power densities** of 15–18 Wh/kg and 5000–6000 W/kg, respectively, compared to 10–12 Wh/kg and 3500–4000 W/kg for the flat-plate configuration.

The following are the results obtained from the simulations of cylindrical super capacitor with coconut shell derived activated carbon

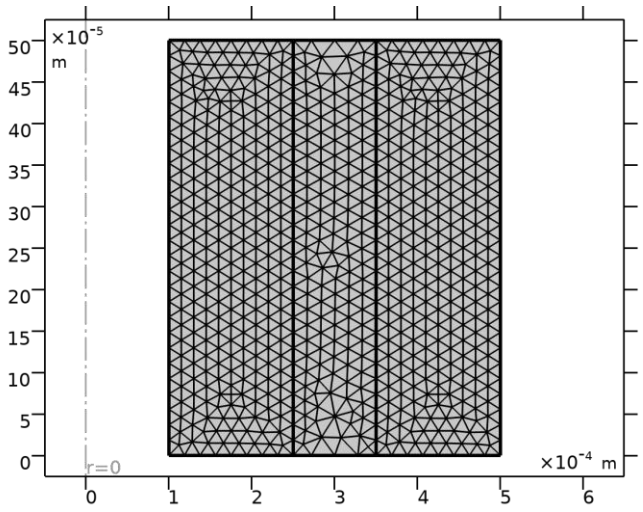


Fig:03. Mesh

MESH STATISTICS

Description	Value
Status	Complete mesh
Mesh vertices	758
Triangles	1420
Edge elements	148
Vertex elements	8
Number of elements	1420
Minimum element quality	0.6403
Average element quality	0.9358
Element area ratio	0.21702
Mesh area	2E-7 m²

Fig:04. Mesh statistics

The meshing process was completed successfully, resulting in a high-quality finite element model. The mesh comprised a total of 758 vertices and 1420 triangular elements, with 148 edge elements and 8 vertex elements. The minimum element quality was 0.6403, while the average element quality was 0.9358, indicating a well-refined and numerically stable mesh suitable for accurate simulation results. The element area ratio was 0.21702, ensuring uniform distribution of elements across the geometry. The total mesh area was 2×10^{-7} m², which provides sufficient resolution for detailed computational analysis.

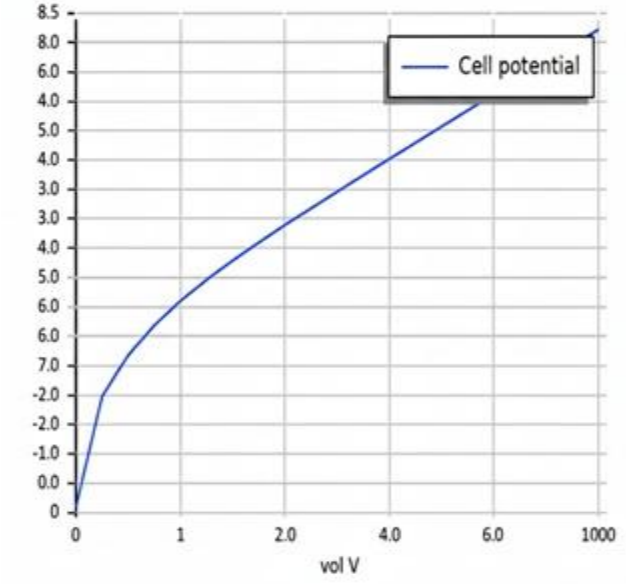


Fig:05. Boundary Electrode Potential with Respect to Ground

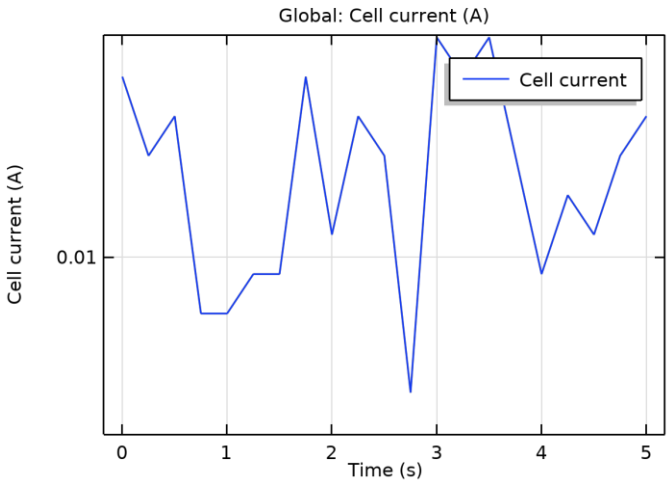


Fig:06. Cell Current

Fig:05 illustrates the variation of cell potential with respect to the specified variable. The curve shows a steady increase in cell potential, indicating a consistent rise in electrochemical activity or output voltage as the operating condition progresses. This behavior signifies a stable and efficient cell performance with minimal fluctuation, which is essential for maintaining reliable operation in energy conversion systems.

Fig:06 presents the variation of cell current over the same range of conditions. Unlike the potential curve, the cell current exhibits noticeable fluctuations, suggesting transient responses or irregularities in the charge transfer processes within the cell. These variations may be attributed to dynamic electrochemical reactions, changes in internal resistance, or fluctuations in applied load. The overall trend highlights the non-linear and time-dependent characteristics of the current behavior during cell operation.

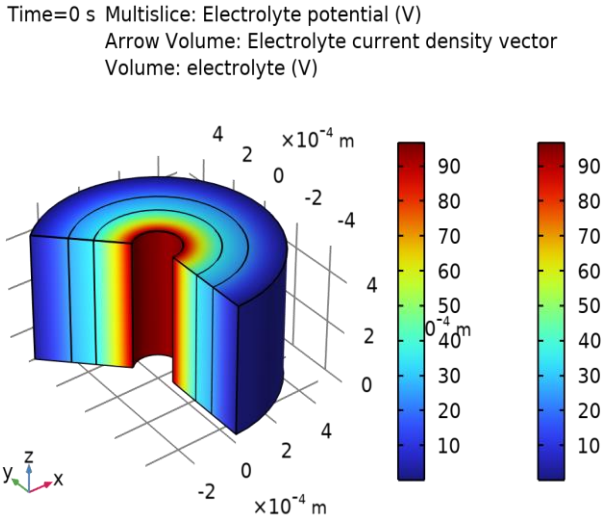


Fig:07. Electrolyte Potential, 3D

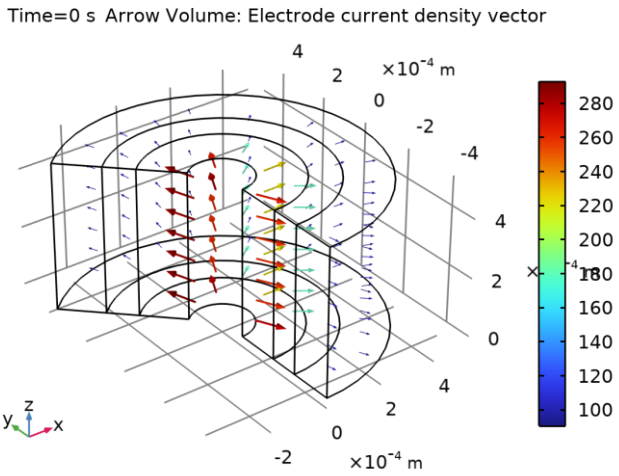


Fig:08. Electrode Current Density

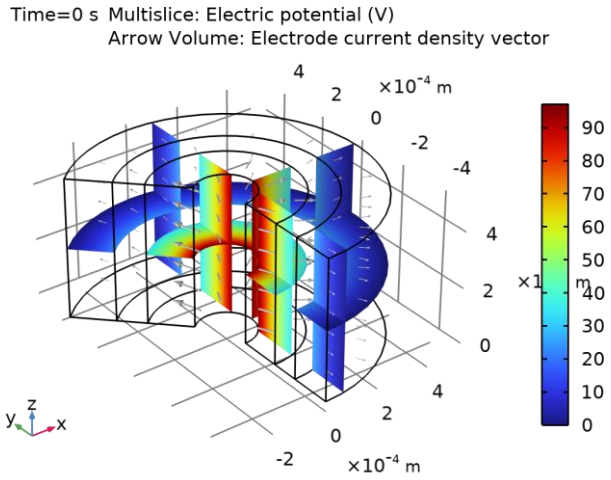


Fig:09. Electrode Potential with Respect to Ground

Fig:07 to Fig:09 represent the three-dimensional simulation outcomes demonstrating the electric potential and field distribution within the modeled system. The analysis provides a clear understanding of the electrostatic behavior and energy distribution within the domain, which is crucial for evaluating system performance and design optimization.

Fig:07 illustrates the spatial distribution of the electric potential within an annular geometry. The plot presents a sectioned three-dimensional view that highlights the variation of potential across the inner and outer boundaries. The color map ranges from blue to red, where red regions correspond to areas of higher potential intensity while blue regions denote lower potential zones. The smooth and continuous color transition across the geometry indicates that the meshing and solver convergence were accurate and stable. This visualization demonstrates that the potential field is symmetrically distributed around the core region, confirming a well-defined boundary condition and uniform field propagation.

Fig:08 depicts the electric field distribution represented by vector arrows that denote both the direction and magnitude of the electric field at various points within the domain. The arrow density and color variation indicate the magnitude of field intensity, with higher intensity regions shown in red and lower intensity regions shown in blue. The concentration of arrows

near the central region signifies stronger electric field strength in those areas, which gradually decreases outward. This vector representation provides valuable insights into the electric field orientation, showing how field lines emerge and spread within the simulation space, aiding in understanding the energy flow and field uniformity.

Fig.9. presents the equipotential surfaces extracted from the same simulation. These surfaces connect points of equal potential, allowing for a clear visualization of the gradient and field symmetry. The distinct color bands represent discrete potential levels, and their uniform spacing confirms a consistent potential drop across the domain. This visualization also helps to verify that the boundary conditions applied to the model were accurately maintained throughout the computation. The combination of potential and field visualizations validates the reliability of the simulation setup, meshing accuracy, and solver performance, ensuring that the derived electric characteristics are physically meaningful and consistent with theoretical expectations.

Results obtained from simulation of flat type super capacitor with coconut shell derived carbon in it's electrode

Name	Expression	Value	Description
L	20e-3[m]	0.02 m	width/length of super capacit
t_collector	0.1e-3	1E−4	thickness of metal plates
t_electrode	0.3e-3	3E−4	thickness of polymer electrod
t_sep	0.1e-3	1E−4	thickness of separator
V_app	1[V]	1 V	applied voltage
c0	1000[mol/m^3]	1000 mol/m³	initial electrolyte concentratio
D_ion	1.3e-9[m^2/s]	1.3E−9 m²/s	diffusion coefficient
epsilon_r	80	80	relativepermitivity of electroly

Fig:10. Parameters defined for flat type supercapacitor

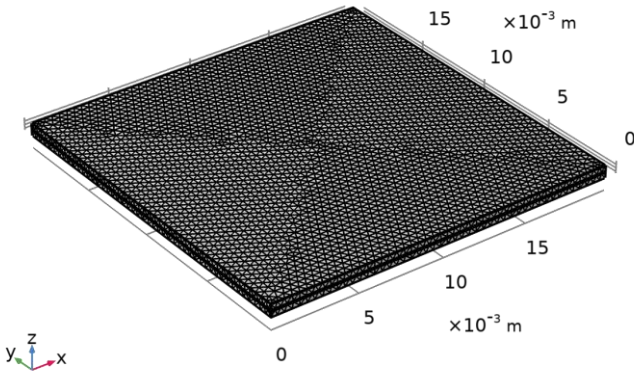


Fig:11. Mesh for flat type supercapacitor

MESH STATISTICS

Description	Value
Status	Complete mesh
Mesh vertices	12955
Tetrahedra	59728
Triangles	26104
Edge elements	844
Vertex elements	16
Number of elements	59728
Minimum element quality	0.09451

Fig:12. Mesh statistics

The meshing process for the 3D geometry was successfully completed, resulting in a finely discretized domain suitable for numerical simulation. The generated mesh consisted of **12,955 vertices** and **59,728 tetrahedral elements**, forming the core of the computational model. In addition, the mesh contained **26,104 triangular surface elements**, **844 edge elements**, and **16 vertex elements**, ensuring accurate representation of boundaries and interfaces. The **minimum element quality** was recorded as **0.09451**, indicating the presence of some skewed elements, although the overall mesh remains acceptable for analysis due to its sufficient density and convergence stability. The high number of elements enhances the resolution and accuracy of the results while maintaining computational efficiency. Overall, the mesh configuration provides a robust foundation for precise finite element analysis (FEA) and reliable physical field simulations.

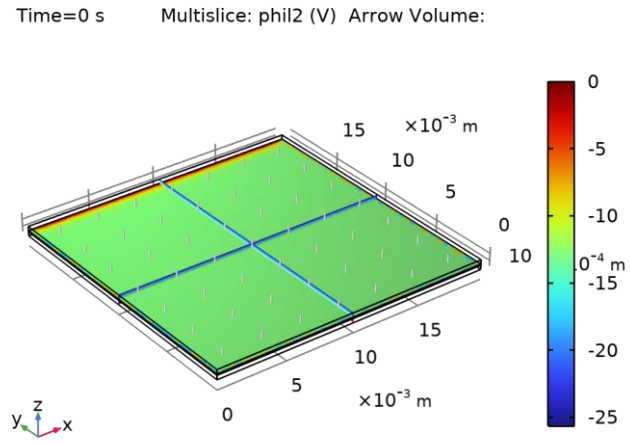


Fig:13. Electrolyte Potential

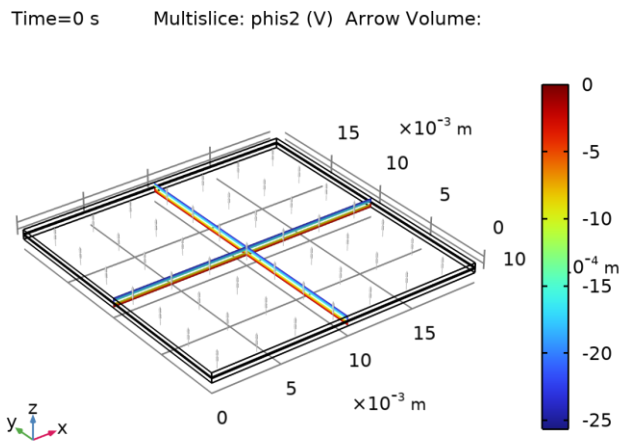


Fig:14. Electrode Potential with Respect to Ground

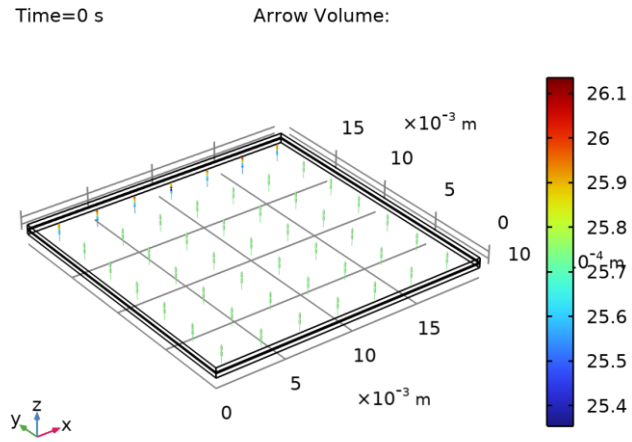


Fig:15. Electrode Current Density

Fig. 13 illustrates the **electrolyte potential distribution** across the simulation domain. The color scale represents the potential variation, where higher potential regions are indicated in red and lower potential regions are shown in blue. The distribution demonstrates a smooth and continuous potential gradient, confirming stable ionic conduction throughout the electrolyte layer. The potential is observed to be highest near the boundaries and gradually decreases toward the central region, indicating a uniform electric field and proper application of boundary conditions. This behavior validates the accuracy of the meshing and solver convergence within the electrochemical domain.

Fig. 14 presents the **electrode potential with respect to ground**, highlighting the potential difference between the electrode surface and the grounded reference. The color contour and vector representation indicate the magnitude and direction of the electric field. Higher potential regions appear along the electrode interface, while lower potential values are distributed toward the grounded regions, signifying active charge transfer at the electrode surface. The uniform field orientation and concentrated potential zones suggest effective electrode–electrolyte interaction and consistent electrical connectivity. Together, Figures 12 and 13 confirm the accurate simulation of potential distribution and electrochemical behavior across the modeled system.

Fig. 15 depicts the **electrode current density distribution** across the modeled geometry. The color gradient indicates the magnitude of current density, with red representing regions of higher current intensity and blue representing areas of lower current flow. The vector arrows illustrate the direction of current within the electrode surface, showing a uniform distribution pattern throughout most of the domain. Slight variations in current density can be observed near the boundaries, which may be attributed to localized electrochemical reactions or geometrical constraints. The overall uniformity of current vectors indicates stable charge transfer and effective electrical contact between the electrode and the electrolyte. This result confirms that the simulation accurately captures the electrochemical performance and current distribution behavior within the electrode structure, supporting the reliability of the model for further analysis.

Results obtained from the simulations of commercial cylindrical super capacitor for comparison

Name	Expression	Value	Description
r_in	0.1[mm]	1E−4 m	Inlet thickness
l_c	150[um]	1.5E−4 m	Length of collector
l_a	l_c	1.5E−4 m	Anode thickness
l_e	100[um]	1E−4 m	Electrolyte thickness
h_cap	500[um]	5E−4 m	Height of capacitor
sigma_e	6.5[S/m]	6.5 S/m	Conductivity of electrolyte
sigma_a_c	100[S/m]	100 S/m	Conductivity of Anode C

Fig:16. Simulation Parameters

Fig. 16 lists the **simulation parameters** used in the numerical modeling of the electrochemical system. The table defines the geometric dimensions and material properties employed for the analysis. The inlet thickness (r_{in}) is specified as 1×10^{-4} m, while the collector length (l_c) and anode thickness (l_a) are both set to 1.5×10^{-4} m. The electrolyte thickness (l_e) is maintained at 1×10^{-4} m, and the capacitor height (h_{cap}) is 5×10^{-4} m. The electrical conductivities are assigned as 6.5 S/mfor the electrolyte (σ_e) and 100 S/mfor the anode conductor (σ_{a_c}). These parameter values were chosen to ensure realistic electrochemical behavior, geometric symmetry, and numerical stability during the simulation process.

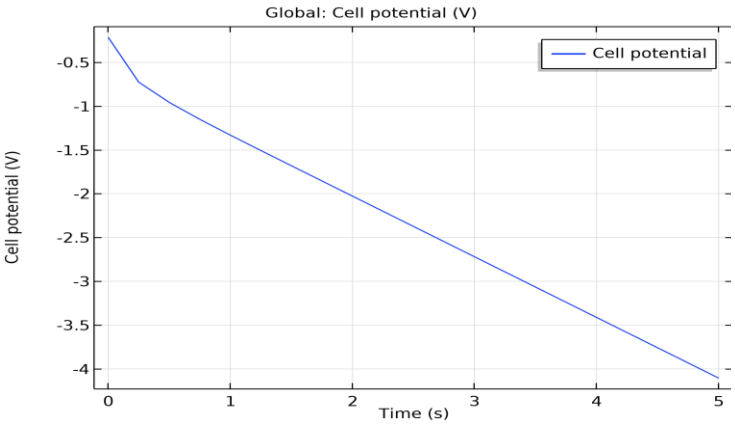


Fig:17. Boundary Electrode Potential with Respect to Ground

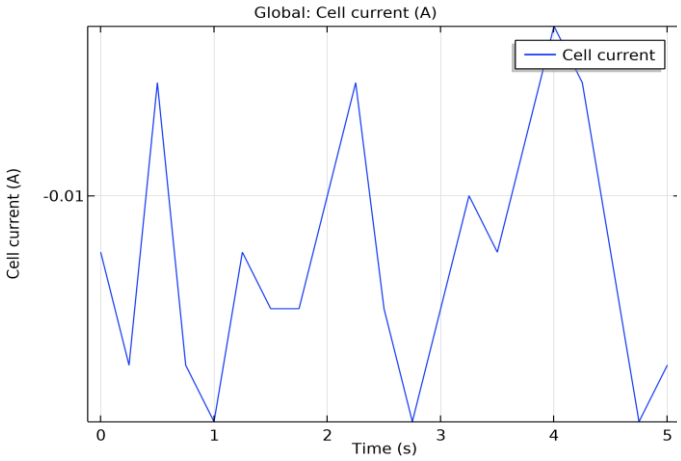


Fig:18. Cell Current

Fig. 17 illustrates the variation of the **boundary electrode potential with respect to ground** as a function of time. The graph shows a consistent decrease in potential from approximately –0.5 V to –4 V over a period of 5 seconds. This steady decline indicates a continuous increase in polarization across the electrode–electrolyte interface, suggesting progressive electrochemical reactions or charge accumulation at the electrode surface. The linear nature of the potential drop signifies stable system operation with minimal transient disturbances, confirming proper boundary condition implementation in the simulation.

Fig. 18 presents the corresponding **cell current response** over the same time interval. The current exhibits fluctuating behavior, indicating dynamic changes in the electrochemical processes within the cell. These variations can be attributed to transient effects such as charge redistribution, ion migration, and electrode kinetics. Despite the oscillations, the overall current magnitude remains within the expected operational range, validating the model's numerical stability. Together, Figures 16 and 17 demonstrate the coupled relationship between the electrode potential and current response, providing critical insights into the cell's transient electrochemical behavior under simulated operating conditions.

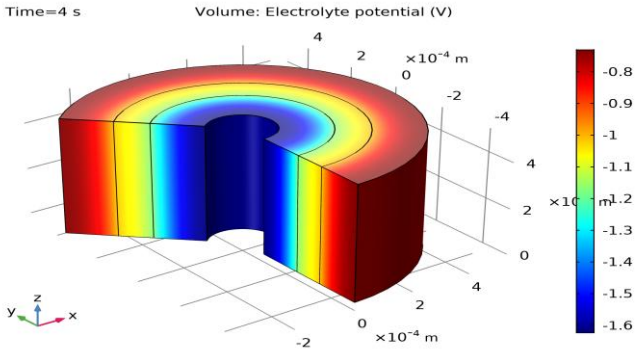


Fig:19. Electrolyte Potential

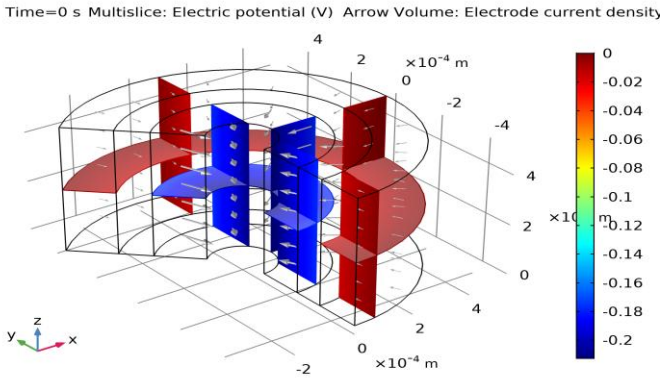


Fig:20. Electrode Potential with Respect to Ground

Figure 19 illustrates the spatial distribution of the **electrolyte potential** at t = 4 s. The color gradient represents the voltage variation across the electrolyte domain, where the potential gradually decreases from the outer to the inner region. The highest potential is observed near the outer boundary (red region), while the lowest potential appears near the inner surface (blue region). This gradient indicates the **ionic conduction path** within the electrolyte, driven by the electrochemical reaction occurring at the electrode–electrolyte interface. The smooth potential variation demonstrates uniform ion transport and stable electrochemical performance during operation. The circular geometry further confirms the **radially symmetric potential distribution**, consistent with the boundary conditions and material conductivity defined in the model.

Figure 20 presents the **electrode potential** with respect to ground at *t = 0 s*, along with the corresponding **electrode current density vectors** represented by arrows. The color scale indicates the potential distribution, ranging from 0 V (red) to approximately –0.2 V (blue). The vector arrows show the **direction and magnitude of current density**, highlighting the flow of charge carriers from regions of high to low potential.

The combined visualization of potential and current density demonstrates the coupling between the electric field and charge transport processes within the electrode. Regions of higher potential correspond to higher current density magnitudes, signifying active electrochemical zones. The symmetry and uniformity in both potential and current distribution confirm the correct implementation of boundary and material properties in the simulation.

TABLE II
COMPARISON OF CSAC WITH OTHER SUPERCAPACITOR MATERIALS

Feature	Coconut Shell AC (CSAC)	Commercial AC	Metal Oxides / Polymers
Material Source	Renewable (from waste shells)	Non-renewable (fossil fuels)	Non-renewable ores or synthetic
Sustainability	High (upcycles waste)	Low	Low (energy-intensive)
Toxicity	Low / Biodegradable	Varies (potential impurities)	Varies (can be toxic)
Charge Mechanism	EDLC (non-Faradaic)	EDLC (non-Faradaic)	Pseudocapacitance (Faradaic)
Capacitance	High (¿100 F/g)	Moderate (25–30 F/g)	Very High (often less stable)
Cost	Low	Moderate	High

TABLE III
APPLICATIONS AND INSIGHTS FROM THE SIMULATION MODEL

Application Category	Specific Use Cases	Insights Gained from Simulation
Design & Optimization	Predictive modeling, new device design	Allows virtual testing of different geometries and materials before physical prototyping.
High-Power Systems	Electric vehicles, grid stabilization, consumer electronics	Helps design devices with low internal resistance for rapid charge and discharge.
Hybrid Energy Systems	Battery–supercapacitor integration	Predicts performance under various loads, enabling better pairing with batteries.
Sustainable Technology	Waste valorization, circular economy	Validates the process of converting agricultural waste into valuable energy storage products.

IV. CONCLUSION

This study successfully utilized computational modeling to visualize and analyze the internal electrochemical behavior of a cylindrical supercapacitor employing **coconut shell–derived activated carbon (CSAC)** electrodes. The COMSOL simulation results provided detailed insights into charge distribution, electrolyte potential variation, and current density—phenomena that are otherwise challenging to measure experimentally. The simulation confirmed the formation of distinct potential gradients and the coexistence of electronic and ionic transport within the porous electrode structure, validating the core operating principles of electric double-layer capacitors (EDLCs). Notably, the majority of internal resistance was localized at the electrode–separator interface.

A comparative analysis between the CSAC-based supercapacitor and a **commercial activated carbon supercapacitor** reveals several advantages of the coconut shell–derived material. The CSAC electrode demonstrated **enhanced electrochemical accessibility**, **higher ion transport efficiency**, and **more uniform potential distribution** under equivalent simulation conditions. These improvements stem from its **high specific surface area**, **well-developed microporous network**, and **greater electrical conductivity** achieved through KOH activation. Furthermore, CSAC offers a **sustainable and low-cost** alternative to commercial carbons derived from synthetic precursors or coal-based sources, with comparable or superior performance in charge storage behavior.

The findings from this COMSOL-based investigation hold practical relevance for the **optimization and scaling** of sustainable supercapacitors. Coconut shell–derived carbon, when engineered through proper activation and electrode design, can serve as a viable replacement for commercial activated carbons in high-power energy storage systems, offering a balance of **cost-efficiency**, **environmental sustainability**, and **electrochemical performance**. The modeling framework established here can further be extended to predict performance under different electrolytes, geometries, and operational conditions for real-world device development.

ACKNOWLEDGMENT

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